Daniel Biasiotto

Computer Science & Artificial Intelligence Student

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ACADEMIC PROFILE

Master's student in Artificial Intelligence with a solid theoretical and methodological background (**110/110 cum Laude in Computer Science BSc**). Research interests focused on the intersection of quantum computing, deterministic multi-agent architectures, and state estimation/control systems. International study experience (Luxembourg, United Kingdom) and professional proficiency in English (**C1**). Active contributor to open-source software for research and scientific automation.

EDUCATION

Master's Degree in Artificial Intelligence
University of Turin — *Sep 2022 – Apr 2027 (Expected)*
- GPA: **27.5 / 30** | **EDISU Merit Scholarship** | Focus: Machine Learning, Multi-Agent Systems, Advanced Statistical Methods.
- **International Mobility:** Erasmus+ Program at the **University of Luxembourg** (Master in Computer Science – AI, 2023/24); academic courses in Paignton (UK).

Bachelor's Degree in Computer Science
University of Turin — *Sep 2019 – Oct 2022*
- Final Grade: **110 / 110 cum Laude** | **EDISU Merit Scholarship**
- Thesis: *"Simulation of Quantum Algorithms using the Q# tool"*
- Supervisor: **Prof. Stefano Berardi** | Co-Supervisor: **Prof. Luca Luigi Paolini**

RESEARCH & OPEN SOURCE PROJECTS

Limen – *Deterministic Multi-Agent Workflow Engine* (github.com/denialbb/limen)
- Designed and developed a deterministic multi-agent orchestration engine in **Go**, formalized as a transactional state machine over SQLite (WAL).
- Implemented concurrent LLM worker isolation via `git worktree`, formal code validation protocols, and an interactive TUI.

AEGIS – *Autonomous Estimation & Guidance Integrated System* (github.com/denialbb/AEGIS)
- Implemented non-linear state filters (12-state Error-State EKF) and Mahony attitude filters for 6-DOF dynamics.
- Pseudo-inverse based control allocation algorithms and Bayesian hyperparameter optimization using **Optuna**.

ISBN-Bridge – *Cryptographic Hardware-Software Bridge* (github.com/denialbb/ISBN-bridge)
- Client-server architecture for secure barcode acquisition: Go protocol, HMAC SHA-256 hashing, and cross-platform native clients (Linux/Windows/iOS/Android).

Open Source Contributions & Academic Tools:
- **huacnlee/omamail** (280+ stars): Fixed the XML CDATA parser for CalDAV protocol responses.
- **markdownreader.koplugin**: Engine for typographic rendering of scientific documents and LaTeX formulas via MicroTeX (Lua/C++).

PROFESSIONAL EXPERIENCE & VOLUNTEERING

Librarian (Universal Civil Service)
Garessio Civic Library — *Sep 2025 – Present*
- Managed documentary heritage; developed analytical scripts and data-cleansing tools for the catalog database (**Biblios**).

International and Erasmus Student Welcome Assistant
University of Turin — *Feb 2023 – Aug 2023*
- Provided linguistic, academic, and administrative support (residency and enrollment procedures) for international and Erasmus students.

First Aid Volunteer & Educator
Italian Red Cross – Mondovì Committee — *Mar 2018 – Dec 2020*
- Achieved First Aid certification (theoretical-practical exam); participated in medical assistance operations and community awareness campaigns.

Software Developer (Unity / C#)
GF Creative Lab — *Sep 2021 – Apr 2022*
- Engineered software architecture and integrated testing and continuous integration pipelines (GitHub Actions).

TECHNICAL SKILLS

Programming:	Go, Python, C++, Rust, C#, Q#, Haskell, Java, Lua, Emacs Lisp (Elisp), SQL
Research Areas & Methods:	Quantum Computing (Simulation), State Estimation (Kalman/EKF), Multi-Agent Systems, Machine Learning
Systems & Tools:	Linux, Git, SQLite WAL, MariaDB, Docker/Kubernetes, Typst, LaTeX
Languages:	Italian (Native), English (**C1 Advanced** – CEFR, international mobility and study), French (Reading)